

KhataAI Software Design Document

Section 3 – Database Design

Purpose: Define every MongoDB collection, fields, indexes, relationships, validation rules, sample document, APIs using it, and business rules.

users

Attribute	Details
Purpose	Authentication and user identity.
Fields	_id,ObjectId; store_id,ObjectId; name; phone(unique); role; password_hash(optional)
Indexes	phone(unique), store_id
Validation	Phone mandatory; verified before login.
Sample	{name:'Suchit', phone:'9876543210', role:'OWNER'}
Used By	/auth/register, /auth/login, /auth/me
Business Rules	One phone per account. OTP verification required.

stores

Attribute	Details
Purpose	Business profile and invoice configuration.
Fields	store_name, gstin, address, phone, logo_url, invoice_prefix, invoice_counter, u
Indexes	gstin(unique optional), owner_id
Validation	GST optional initially; invoice prefix required.
Sample	{store_name:'Shree Ram Traders', invoice_prefix:'INV'}
Used By	/store/profile
Business Rules	Invoice counter increments atomically.

customers

Attribute	Details
Purpose	Customer directory and credit.
Fields	name, phone, gstin, address, outstanding_balance, credit_limit, status, store_i
Indexes	(store_id,phone unique), (store_id,name)
Validation	Outstanding cannot be manually negative.
Sample	{name:'Ramesh', outstanding_balance:500}
Used By	/customers/*
Business Rules	Soft delete only. Credit warnings above limit.

products

Attribute	Details
Purpose	Inventory catalog.
Fields	name, aliases[], sku, price, gst_rate, stock, low_stock_threshold, active, store_id
Indexes	(store_id,name), sku(unique/store)
Validation	Price>=0, Stock>=0 preferred.
Sample	{name:'Coca Cola 250ml', aliases:['Coke']}
Used By	/products/*, voice processing
Business Rules	AI matches aliases; backend calculates totals.

transactions

Attribute	Details
Purpose	Immutable sales ledger.
Fields	invoice_number, customer_id, items[], subtotal,gst_amount,grand_total,payment_status
Indexes	invoice_number(unique), customer_id+created_at
Validation	Totals calculated only by backend.
Sample	{invoice:'INV-2026-0001', grand_total:118}
Used By	/transactions/*
Business Rules	Never delete. Reverse using reversal transaction.

voice_transcripts

Attribute	Details
Purpose	Store AI transcript and extraction.
Fields	audio_url, transcript, ai_output, confidence,status
Indexes	store_id, created_at
Validation	Status=PENDING/CONFIRMED/REJECTED
Sample	{transcript:'5 Coke', confidence:0.94}
Used By	/voice/process,/voice/confirm
Business Rules	Draft only until user confirms.

expenses

Attribute	Details
Purpose	Business expenses.
Fields	category, amount, note, date, store_id
Indexes	store_id,date
Validation	Amount>0
Sample	{category:'Transport',amount:500}

Used By	/expenses
Business Rules	Included in analytics.

notifications

Attribute	Details
Purpose	Alerts.
Fields	title,message,type,is_read,store_id
Indexes	store_id,is_read
Validation	Read defaults false.
Sample	{title:'Low Stock'}
Used By	/notifications
Business Rules	Generated by background tasks.

audit_logs

Attribute	Details
Purpose	Track sensitive operations.
Fields	user_id,action,entity,old_value,new_value,time
Indexes	store_id,time
Validation	Append only.
Sample	{action:'UPDATE_PRODUCT'}
Used By	Admin/Audit
Business Rules	Never modify/delete.

Collection Relationships

Store

- Users
- Customers
- Products
- Transactions
- Expenses
- Notifications
- VoiceTranscripts
- AuditLogs

Input / Output Example

Voice Input: 'Ramesh ne 5 Coke ane 2 Biscuit lidha'

AI Output (Draft): {customer:'Ramesh',
items:[{product:'Coke',qty:5},{product:'Biscuit',qty:2}]}

Backend: Fetch prices + GST from Products collection → Calculate totals → Create Transaction → Update Customer outstanding → Reduce Product stock → Generate Invoice.

Production Rules

- Every collection except users contains store_id.
- No financial data is deleted.
- Soft delete for customers/products.
- Invoice numbers generated atomically.
- Redis stores OTP and cache only.
- MongoDB is the source of truth.